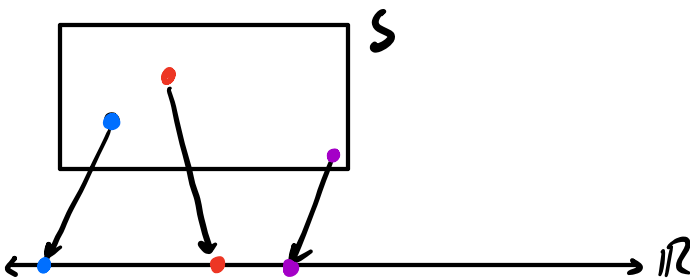


Random Variables:

Our discussion of probability has one important caveat - the sample space S can be anything. E.g. Letters, words, colors, cards, etc.

However, recall from our first week of classes that we are interested in numerical variables. Hence, we must define a mapping from the sample space S to the real numbers.

Def. (Random Variable): A random variable $X(\cdot)$ is a function mapping S (the sample space of an experiment) to the real numbers.



Notation: $\omega \in S$ will refer to a single outcome in the sample space.

$X(\omega) = x$ will be the real number associated with outcome ω .

(Very Important) Note: Capitalized " X " refers to the random variable, without an associated outcome of the experiment. In contrast, the lowercase " x " is a realization of the random variable " X ".

Realizations are just real numbers - without any uncertainty.

For a random variable (r.v.) X , the set of all possible values it can take is called the **range** of X .

(Not to be confused with the range of some numbers, used as a measure of variability)

Example:

Experiment: Flip 2 coins, each is equally likely to land on H vs. T
 $S = \{(HH), (HT), (TH), (TT)\}$, all w/ equal probability.

R.V. X :

ω	HH	HT	TH	TT
$X(\omega)$	2	1	1	0

|| Note: X "loses" some info. from the original experiment.
 $X(HT) = X(TH) = 1$

I.e. $X(\omega) = \text{"# of H's in } \omega\text{"}$

We know how likely each ω is. So, the natural thing to do, having defined X , is to compute how likely X is to take a certain value.

I.e. we want to compute $\mathbb{P}(\{\text{outcomes } \omega \in S \text{ for which } X(\omega) = x\})$ for different values of x . In short, we write this as $\mathbb{P}(X=x)$

Note: $\{\omega \in S \text{ for which } X(\omega) = x\}$, in the above, is a well-defined event! I.e. a subset of S .

We compute:

x	0	1	2	anything else
$\mathbb{P}(X=x)$	$1/4$	$1/4 + 1/4$	$1/4$	0

} Called the distribution of X .

Goal: We want to define and analyze the distributions of different random variables.

Case I: Discrete R.V.'s

If the range of a random variable X is either finite or countable, then X is said to be a **discrete r.v.**

In the example above, $\text{range}(X) = \{0, 1, 2\}$, which has 3 items, is finite, and hence this X is discrete.

If Necessary: Review of Set Cardinality

For some set A , A is:

- **Finite** if $|A| = N$, for some integer N . e.g. $\{1, 2, \dots, 1000\}$
- **Countably Infinite** if we can write it as $A = \{x_1, x_2, x_3, x_4, \dots\}$, but $|A| = \infty$.
I.e. every item can be 1-to-1 associated with a natural #. e.g. $\mathbb{Z}$
- **Uncountably Infinite** if $|A| = \infty$, but A is not countably infinite. e.g. $\mathbb{R}$

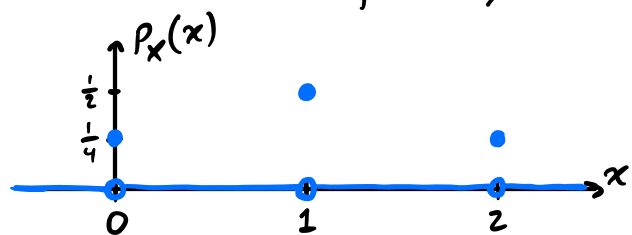
We can describe the distribution of a discrete r.v. by specifying the probability that it takes each possible value.

Def. (Probability Mass Function): For a r.v. X , its **probability mass function (pmf)** is a function $p_X(\cdot)$ such that:

$$p_X(x) = \mathbb{P}(X=x), \text{ for all } x \in \mathbb{R}.$$

Note: In the above example, $\star$ is the pmf of X .

If we wanted to plot it, we would draw:



Reminder: Open circle means "up to but not including".

Closed circle means "only at".

Properties of PMF:

- $P_X(x) \in [0, 1]$ for all x (why?)

- $\sum_{x \in \text{range}(X)} P_X(x) = 1$ (why?)

A discrete r.v. is fully characterized by its pmf. However, it is also fully characterized by the following:

Def. (Cumulative Distribution Function): The cumulative distribution function (cdf), $F_X(\cdot)$, of a r.v. X with pmf $p_X(\cdot)$ is defined as:

$$F_X(x) = \mathbb{P}(X \leq x) = \sum_{y \leq x} p_X(y) = \sum_{y \leq x} \mathbb{P}(X=y)$$

why does this equal $\mathbb{P}(X \leq x)$?

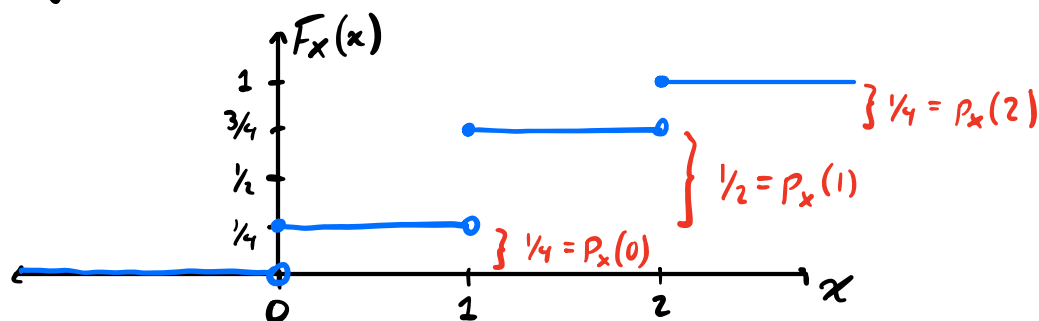
Ex: Recall the pmf from an earlier example

x	0	1	2	anything else
$p_X(x)$	$1/4$	$1/2$	$1/4$	0

The CDF of this r.v. would be:

x	$x < 0$	$0 \leq x < 1$	$1 \leq x < 2$	$2 \leq x$
$F_X(x)$	0	$1/4$	$1/4 + 1/2$ " $3/4$	$1/4 + 1/2 + 1/4$ " 1

Plotting it looks like:



Note: Size of the jump at any point x is equal to $P_X(x)$!

Properties of CDF:

- $0 \leq F_X(x) \leq 1$ (why?)
- If $a \leq b$, then $F(a) \leq F(b)$ [Non-decreasing]

Pf: $A_1 = \{\omega \in S \text{ s.t. } X(\omega) \leq a\}$
 $A_2 = \{\omega \in S \text{ s.t. } a < X(\omega) \leq b\}$

Note: $A_1 \cap A_2 = \emptyset$, and $A_1 \cup A_2 = \{\omega \in S \text{ s.t. } X(\omega) \leq b\}$

Use Axiom 3, and: i) $P(A_1) = P(X \leq a)$, ii) $P(A_2) = P(a < X \leq b)$,

iii) $P(A_1 \cup A_2) = P(X \leq b)$, and iv) Fact that $P(A_2) \geq 0$.

- As $x \rightarrow \infty$, $F_X(x) \rightarrow 1$ (why?)

Expected Value:

When we were given some numerical variables (e.g. from a sample), we asked about a measure of location. Now, we build a measure of location for a random variable!

Def. (Expected Value - Discrete): Let X be a discrete r.v. with pmf $p_X(\cdot)$. Then, the **expected value** or **expectation** of X is

$$E[X] = \sum_{x \in \text{range}(X)} x \cdot p_X(x).$$

We often denote $\mu_X = E[X]$, and μ_X is called the **mean** of X .

Interpretation: $E[X]$ is a probability-weighted average of the possible values X can take.

Connection to Descriptive Statistics:

Let X have pmf: $p_X(x) = \begin{cases} 1/4 & \text{if } x=0 \text{ or } 2 \\ 1/2 & \text{if } x=1 \\ 0 & \text{o/w.} \end{cases}$

$$\text{Then: } E[X] = \frac{1}{4} \cdot 0 + \frac{1}{2} \cdot 1 + \frac{1}{4} \cdot 2 = \frac{1}{2}$$

Now imagine a sample of size 4, where the values of some variable are (0, 1, 1, 2). Sample mean is equal to: $\frac{1}{4} (0 + 1 + 1 + 2) = \frac{1}{2}$

These are the same because: $\frac{\text{\# of 1's in sample}}{\text{sample size}} = \frac{2}{4} = p_X(1)$,

and the same holds if we replace "1" with "0" or "2".

$\leadsto$ Frequency in Sample $\approx$ PMF of R.V.

Example:

Experiment: Rolling 6-sided die, w/ equally likely outcomes.

$$S = \{1, 2, 3, 4, 5, 6\}$$

$X(\omega) = \omega$. Its pmf is:

x	1	2	3	4	5	6
$P_X(x)$	$\frac{1}{6}$	$\frac{1}{6}$	$\frac{1}{6}$	$\frac{1}{6}$	$\frac{1}{6}$	$\frac{1}{6}$

Compute $E[X]$:

$$\begin{aligned} E[X] &= \frac{1}{6} \cdot 1 + \frac{1}{6} \cdot 2 + \frac{1}{6} \cdot 3 + \frac{1}{6} \cdot 4 + \frac{1}{6} \cdot 5 + \frac{1}{6} \cdot 6 \\ &= \frac{1}{6} (1 + 2 + 3 + 4 + 5 + 6) \end{aligned}$$

$$\underline{E[X] = \frac{21}{6} = 3.5}$$

We can also compute the expectation of a function of a random variable.

If X is a r.v. with pmf $P_X(\cdot)$, and $g(\cdot)$ is some function,

then:

$$E[g(X)] = \sum_{x \in \text{range}(X)} g(x) \cdot P_X(x).$$

Note: In general, $E[g(X)] \neq g(E[X])$.

An example: X has pmf $P_X(x) = \begin{cases} \frac{1}{2} & \text{if } x=1 \text{ or } x=-1 \\ 0 & \text{o/w} \end{cases}$

Let $g(x) = x^2$.

$$\bullet g(E[X]) = g(0) = \underline{0}$$

$$\bullet E[g(X)] = \frac{1}{2} \cdot 1^2 + \frac{1}{2} \cdot (-1)^2 = \underline{1}$$

However, there is an extremely useful property of the expectation.

Prop. (Linearity of Expectation): Let X be a discrete r.v. Then, for any constants a, b , we have $E[aX+b] = a \cdot E[X] + b$.

So, if $g(x) = a \cdot x + b$, then $E[g(X)] = g(E[X])$.

There is one very special function whose expectation gives us a measure of the variability of a random variable.

Def. (Variance): If X is a discrete r.v. with pmf $p_X(\cdot)$, then the variance of X is: $\text{Var}(X) = \sigma_X^2 = E[(X - E[X])^2]$.

The standard deviation of X is $\sigma_X = \sqrt{\text{Var}(X)}$

Properties:

- $\text{Var}(X) = E[X^2] - E[X]^2$

Pf: $\text{Var}(X) = E[(X - E[X])^2]$

$$= E[X^2 - 2 \cdot E[X] \cdot X + E[X]^2]$$

} Expand the square

$$= E[X^2] - 2 \cdot E[E[X] \cdot X] + E[E[X]^2]$$

} Linearity of Expectation

These are constants - not random
 $\leadsto$ We can pull them out by linearity of expectation

$$= E[X^2] - 2 \cdot E[X] \cdot E[X] + E[X^2] \cdot E[1]$$

$$\text{Var}(X) = \underline{E[X^2] - E[X]^2} \quad \blacksquare$$

• If $Y = a \cdot X + b$, then $\text{Var}(Y) = a^2 \cdot \text{Var}(X)$

Example:

Let X be a r.v. w/ pmf given by:

x	0	1	2	o/w
$P_X(x)$	$\frac{1}{4}$	$\frac{1}{2}$	$\frac{1}{4}$	0

Compute $\text{Var}(X)$:

$$\begin{aligned}\text{Var}(X) &= E[(X - E[X])^2] \\ &= E[(X - 1)^2] \quad \left. \begin{array}{l} \text{ } \end{array} \right\} E[X] = 1 \\ &= \frac{1}{4}(0-1)^2 + \frac{1}{2}(1-1)^2 + \frac{1}{4}(2-1)^2 \\ &= \frac{1}{4} \cdot 1 + \frac{1}{2} \cdot 0 + \frac{1}{4} \cdot 1\end{aligned}$$

$$\underline{\underline{\text{Var}(X) = \frac{1}{2}}}$$
